Data 101

CAPTSONE PROJECT_LINH PHAN

CANDY SUPPLY PLANNING

Project Introduction

Candy Supply provides fast delivery of candy for class room instruction in High Schools and Colleges across the United States. Their Supply Chain is critical in the stressful times of student life.

Studies that show that candy shared with friends during Mid-Terms, Finals, and Late-Night study sessions boosts the mood and productivity of the students forming life-long friendships and greater demand for candy!

Candy Supply's Conundrum

Answers to these Specific Questions are needed to ensure the supply of candy is reliable.

- How many days should be added to the ordering cycle to ensure that candy inventories are maintained? (hint: More than the average days late)
- Are the order costs for candy changing and what is the forecasted changes for the 12 months beyond the end of the data recorded?
- Which product has the highest Average Days Late?
- Which Supplier has the highest Average Days Late?
- Are the Days Late for each order changing over time, and what is the forecasted changes for the 12 months beyond the end of the data recorded?

Ordering Cycle Changes

(use Microsoft Excel for this section)

- Specific Question

How many days should be added to the ordering cycle to ensure that candy inventories are maintained? (hint: More than the average days late)

- Obtain the Data

Calculate the days between Actual Delivery and Planned Delivery (if negative set the value to zero)

- Analyze the Data

Create a statistical summary of Days Late

- Report the Results

Paste the Statistical summary on the next slide

Ordering Cycle Changes

- WE ANALYZE THE DELIVERY DELAYS FROM THE DATA TO DETERMINE HOW MANY DAYS SHOULD BE ADDED TO THE ORDERING CYCLE TO MAINTAIN CANDY INVENTORIES. KEY STATISTICS INCLUDE:
- MEAN DAYS LATE: 15.39
- 95TH PERCENTILE: 41.05 DAYS
- USING THE 95TH PERCENTILE ENSURES THAT 95% OF ORDERS WILL ARRIVE WITHIN THE ADJUSTED CYCLE, BALANCING RISK AND PRACTICALITY. ADDING **41 DAYS** ACCOUNTS FOR MOST DELAYS WHILE AVOIDING EXCESSIVE BUFFER FROM THE MAXIMUM (51 DAYS).

Statistic	
Mean	15.39
Median	14.00
Standard Deviation	13.14
Min	0.00
Max	51.00
25th percentile	3.00
75th percentile	24.00
90th percentile	36.00
95th percentile	41.05

Days Late	
Mean	15.385
Standard Error	0.928815
Median	14
Mode	0
Standard Deviatior	13.13543
Sample Variance	172.5395
Kurtosis	-0.398239
Skewness	0.629273
Range	51
Minimum	0
Maximum	51
Sum	3077
Count	200

Order Costs Changes

(use Microsoft Excel for this section)

- Specific Question

Are the order costs for candy changing and what is the forecasted changes for the 12 months beyond the end of the data recorded?

- Obtain the Data

Locate the columns and data for Order costs and Actual Delivery Date

- Analyze the Data

Create a line graph of the Order Cost with Actual Delivery Date on the Independent axis

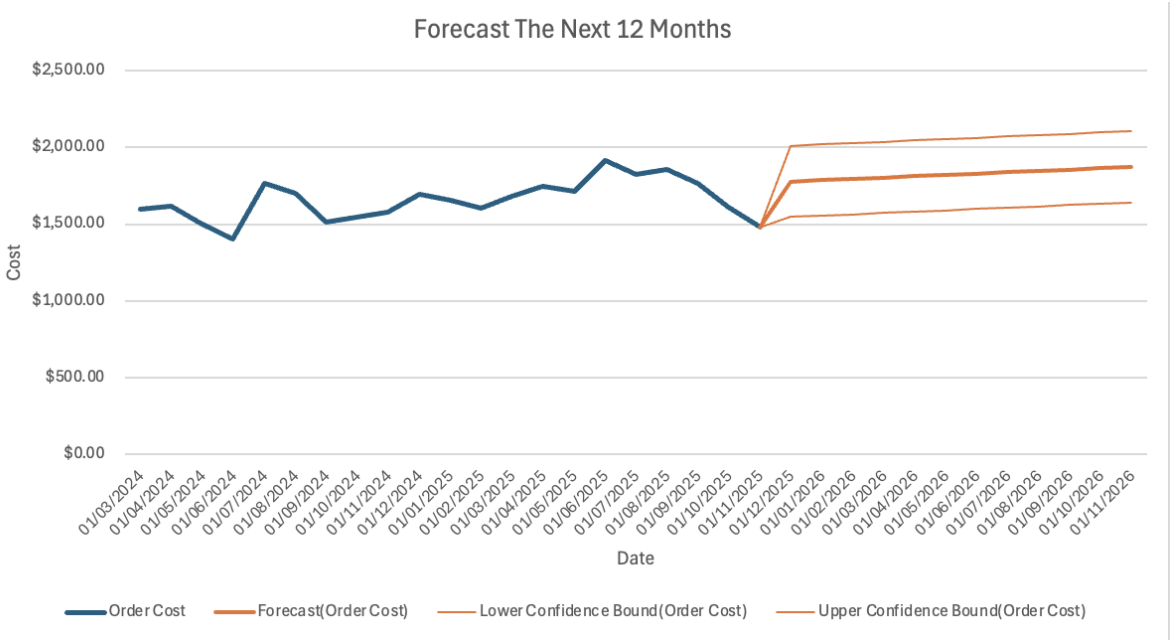
Perform a regression of the Order Cost relative to Actual Delivery Date

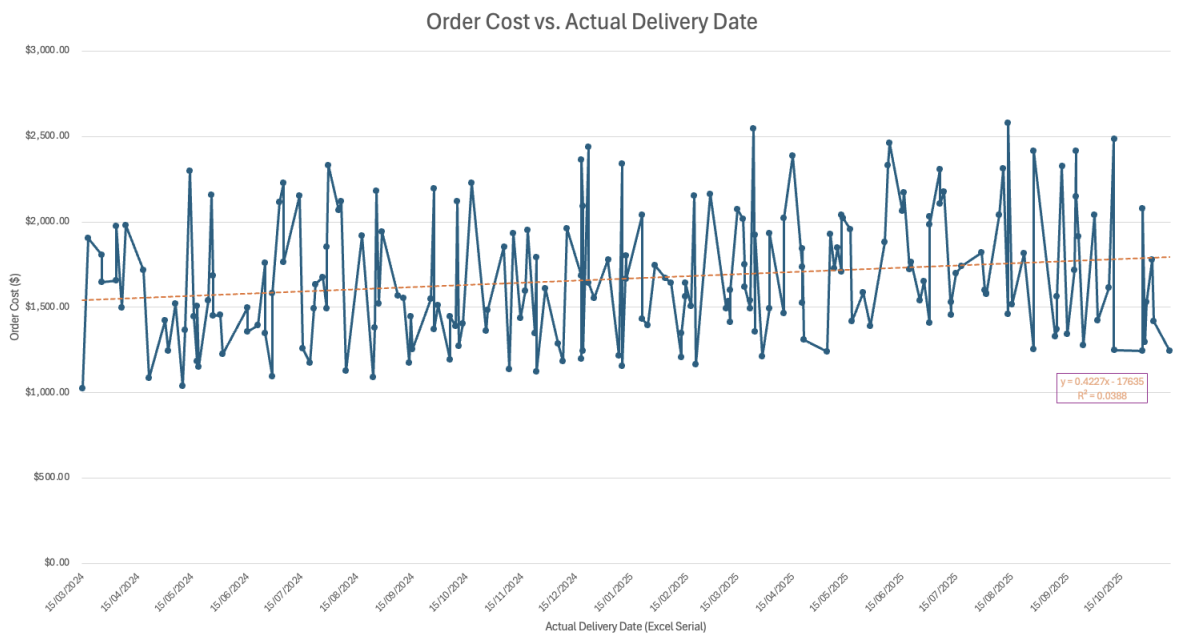
Forecast the following 12 months of Order Costs

- Report the Results

Paste the Line graph, regression results, and forecast results on the next slide.

Order Costs Changes





Order Costs Changes

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.19685161
R Square	0.03875056
Adjusted R Sq	0.03389576
Standard Erro	359.279639
Observations	200

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	1030320.3	1030320.3	7.98191402	0.00520859
Residual	198	25558208.1	129081.859		
Total	199	26588528.4			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-17635.1938	6833.14911	-2.58082965	0.01057946	-31110.2833	-4160.10432	-31110.2833	-4160.10432
Actual Deliver	0.42268309	0.14961025	2.82522814	0.00520859	0.12764906	0.71771712	0.12764906	0.71771712

- Historically, candy order costs fluctuated with individual orders between **$2 and $15** and total costs from **$1,400,000 to $1,700,000**. A weak upward trend of **$0.42 per day** explains just **3.88%** of the variability. Over the next 12 months, total costs are expected to rise **6.67%**, from **$1,500,000 to $1,600,000**, staying within the historical range.

Average Days Late by Product

(use PowerBi for this section)

- Specific Question

Which product has the highest Average Days Late?

- Obtain the Data

Locate the columns and data for Days Late and Product (hint: Days Late were calculated earlier in the project)

- Analyze the Data

Create a table showing the Count of Orders, and Average Days Late by Product

- Report the Results

Paste the table on the next slide

Average Days Late by Product

Product	Average of Days Late	Count of Order Number
Dark Chocolate	16.62	97
Kit Kats	15.19	57
Rollos	14.00	17
Skittles	12.45	29
Total	**15.39**	**200**

IN SUMMARY, THE CHART INDICATES THAT DARK CHOCOLATE EXPERIENCES THE MOST SIGNIFICANT DELIVERY DELAYS (16.62 DAYS), WHILE SKITTLES HAS THE BEST PERFORMANCE (12.45 DAYS), WITH AN OVERALL AVERAGE DELAY OF 15.39 DAYS ACROSS 200 ORDERS.

Average Days late by Supplier

(use Tableau for this section)

- Specific Question

Which Supplier has the highest Average Days Late?

- Obtain the Data

Locate the columns and data for Days Late and Supplier

- Analyze the Data

Create a table showing the Count of Orders, and Average Days Late by Product

- Report the Results

Paste the table on the next slide

Average Days late by Supplier

THE SUPPLIER WITH THE HIGHEST AVERAGE DAYS LATE IS KAREN'S KATS, WITH AN AVERAGE OF **16.58 DAYS LATE** ACROSS ALL THEIR ORDERS.

Supplier	Avg. Days Late	Count of Order Number
Karen's Kats	16.58	52.00
Harvey's Hersheys	16.10	103.00
Ron's Rollos	13.40	25.00
Sean's Skittles	11.10	20.00

Average Days Late Changes over time

(use Tableau for this section)

- Specific Question

Are the Days Late for each order changing over time, and what is the forecasted changes for the 12 months beyond the end of the data recorded?

- Obtain the Data

Locate the columns and data for Days Late and Actual Delivery Date

- Analyze the Data

Create a line graph of the Order Cost with Actual Delivery Date on the Independent axis

Perform a regression of the Order Cost relative to Actual Delivery Date

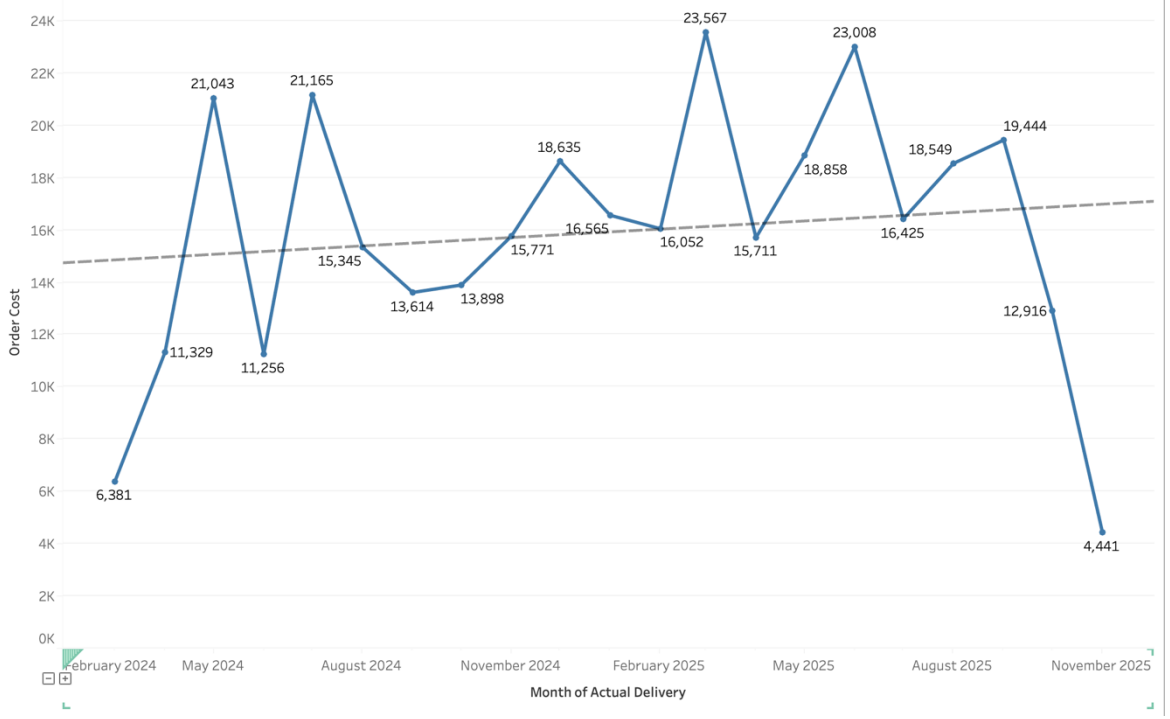
Forecast the following 12 months of Order Costs

- Report the Results

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Average Days Late Changes over time

Line Graph



Describe Trend Model

Trend Lines Model

A linear trend model is computed for sum of Order Cost given Actual Delivery Month.

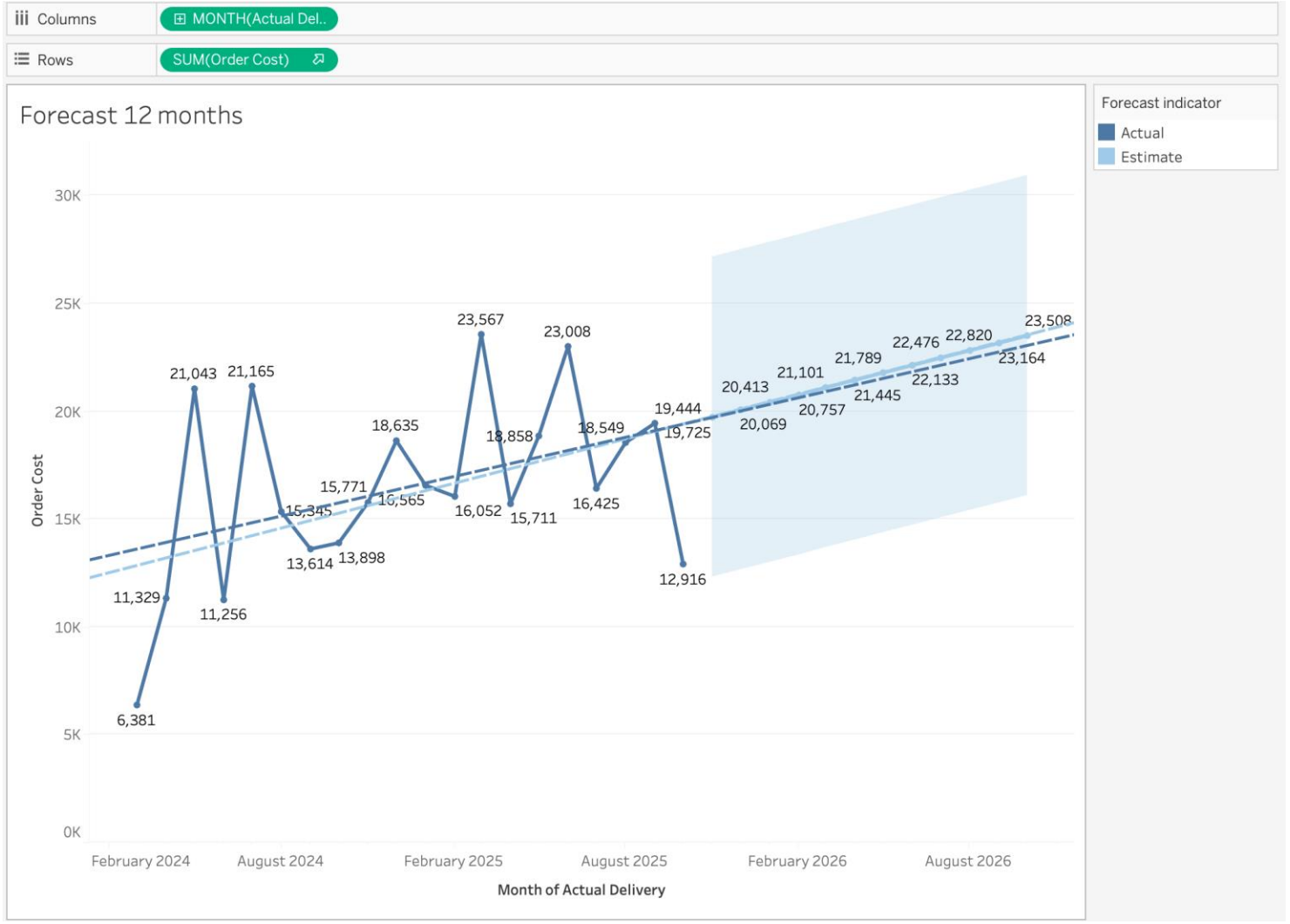
Model formula:	(Month of Actual Delivery + intercept)
Number of modeled observations:	21
Number of filtered observations:	0
Model degrees of freedom:	2
Residual degrees of freedom (DF):	19
SSE (sum squared error):	4.74184e+08
MSE (mean squared error):	2.49571e+07
R-Squared:	0.0180389
Standard error:	4995.7
p-value (significance):	0.561622

Individual trend lines:

Panes		Line		Coefficients				
Row	Column	p-value	DF	Term	Value	StdErr	t-value	p-value
Order Cost	Month of Actual Delivery	0.561622	19	Month of Actual Delivery	3.49335	5.913	0.590792	0.561622
				intercept	-143593	269973	-0.531879	0.600974

Average Days Late changes over time

- ARE DAYS LATE CHANGING OVER TIME?
  DAYS LATE FLUCTUATE (0 TO 51) WITH A SLIGHT DOWNWARD TREND (~4.38 DAYS/YEAR), BUT THE TREND IS WEAK (R-SQUARED ≈ 0.01).
- FORECASTED CHANGES FOR THE NEXT 12 MONTHS?
  AVERAGE DAYS LATE ARE FORECASTED TO DECREASE FROM ~12.8 (DECEMBER 2025) TO ~11.7 (NOVEMBER 2026), A REDUCTION OF ~1.1 DAYS (8.6%), WITH A CONFIDENCE INTERVAL OF ±5 DAYS.



THANK YOU FOR WATCHING MY PRESENTATION.